STEFFI MARY VARGHESE

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WORK EXPERIENCE

Research Intern | Indian Institute of Science Education and Research, Mohali *Jul 2024 - Apr 2026*

- Cardiovascular Risk Prediction: Developed real-time risk prediction models using Random Forest and XGBoost, applying SHAP and Gini-based analysis to improve model interpretability and support transparent risk assessment.
- Baggage Detection & Counting: Developed a YOLO-based real-time computer vision system for baggage detection and counting, delivering the project for Suraj Informatics to support logistics automation and improve operational accuracy.

Analytics Intern | State Bank of India *Jun 2025 - Jul 2025*

- Analyzed customer application data to identify opportunities for support automation, using TF-IDF and Sentence Transformers for query clustering and evaluation.
- Built an AI-powered support automation tool integrating a custom UI, FAISS semantic search, and LLMs to improve query handling and response generation.

Data Engineer Intern | E-Cosys Consultancy *Jan 2025 - May 2025*

- Analyzed Mitsco Finance data and developed KPI dashboards using Excel and Power BI to support data-driven business insights.
- Conducted competitor benchmarking and UX analysis and co-developed Career Copilot using Figma, React.js, JIRA, Cypress, and LLMs to support digital strategy and product improvements.

Software Intern | Concerto Software and Systems *Jun 2024 - Jul 2024*

- Developed Oracle SQL scripts to enhance backend fraud detection and improve the precision of real-time fraud alerts.
- Analyzed historical fraud data to identify attack patterns and collaborated with the PRM team on data-driven enhancements to fraud detection and risk models.

UI/UX Intern | Ideaplots *Nov 2023 - Jul 2024*

- Designed user-centric web interfaces using HTML, CSS, and JavaScript based on client requirements, focusing on usability and responsive design.
- Created Figma wireframes and prototypes, collaborating with clients on UI/UX requirements and supporting advertising and branding design.

PROJECTS

MediSage *Jul 2024 - Jul 2025*

- Developed cardiovascular risk prediction models using Python and XGBoost, incorporating SHAP/LIME explainability techniques to interpret model predictions and identify influential features.
- Built a Next.js analytics dashboard to visualize risk predictions and model insights, connecting machine learning outputs with an interactive decision-support interface.

E-Commerce Support Assistant *Jul 2025 - Aug 2025*

- Built an AI-powered query assistant using Python, FAISS, semantic search, and LLM-based response generation to automate customer query handling and information retrieval.
- Engineered a retrieval workflow combining vector similarity search and LLM-generated responses, enabling relevant information to be identified and transformed into automated support responses.

ASD Prediction using Machine Learning *Jul 2025*

- Developed a machine learning pipeline for Autism Spectrum Disorder prediction using behavioral and demographic features, implementing data cleaning, missing-value imputation, categorical encoding, and feature preparation.
- Trained and benchmarked Logistic Regression, SVM, Decision Tree, Random Forest, and XGBoost models across 70:30 and 80:20 validation splits, comparing accuracy, precision, recall, specificity, F1-score, and ROC-AUC.

EDUCATION

Masters of Data Science and Decisions (Specialization in Computational Data Science) *Sep 2026 - Present*
University of New South Wales, Sydney

Bachelor of Engineering in Computer Engineering (Honours in Data Science) *Oct 2022 - Jun 2026*
Fr. Conceicao Rodrigues Institute of Technology, Navi Mumbai

- GPA : 9.17/10

SKILLS

Programming: Python, C/C++, Java, JavaScript
Machine Learning: Scikit-learn, XGBoost, NLP, NLTK, PySpark, YOLO, OpenCV
Web Development: HTML, CSS, Django, React.js, Cypress
Databases: MySQL, Oracle SQL, Microsoft SQL
Data Processing: NumPy, Pandas
Data Visualization: Matplotlib, Seaborn, Plotly, Power BI, Excel
Systems & Tools: AWS, Git/GitHub, VS Code, Jupyter, Google Colab, Linux, JIRA
UI/Design: Figma, Canva